Lab 04: Router Configuration & Subnetting

Switch0 (2960-24TT) · Router0 (2911) · PC0 · PC1 · PC2

Lab #04	**Topic:** Subnetting & Routing	**Tool:** Cisco Packet Tracer	**Cert:** CompTIA Network+

1. Lab Objectives

- Connect Switch0 (2960-24TT) to Router0 (2911) via Fa0/3 to Gig0/0.
- Assign IP address and subnet mask to both Router0 GigabitEthernet interfaces.
- Configure static IPs, subnet masks and gateways on PC0 and PC1 (on switch subnet).
- Connect PC2 directly to Router0 Gig0/1 as a separate /24 subnet.
- Verify full connectivity between all devices using ping and show commands.

2. Network Topology

Switch0 (2960-24TT) connects PC0 on Fa0/1 and PC1 on Fa0/2. Switch0 uplinks to Router0 on Fa0/3 → Gig0/0. PC2 plugs directly into Router0 Gig0/1. This creates two /24 subnets: 192.168.1.0/24 for the switch LAN and 192.168.2.0/24 for PC2.

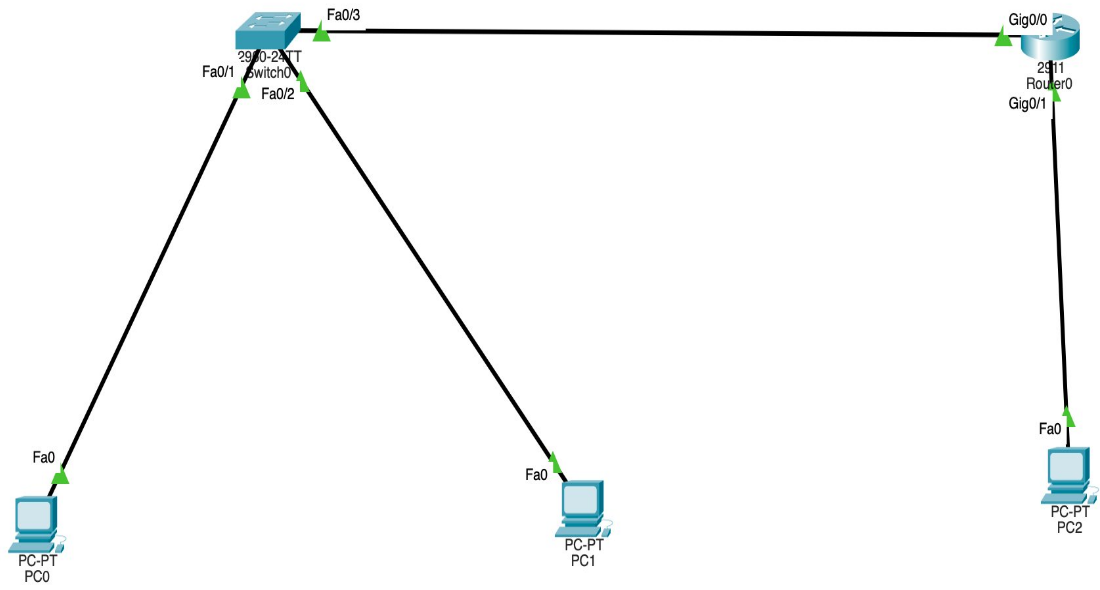

Figure 1 – Switch0 (PC0/PC1) connected to Router0 Gig0/0; PC2 on Router0 Gig0/1

3. IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Role

Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0	—	LAN gateway
Router0	GigabitEthernet0/1	192.168.2.1	255.255.255.0	—	PC2 gateway
Switch0	—	—	—	—	Layer 2 only
PC0	Fa0	192.168.1.10	255.255.255.0	192.168.1.1	Client LAN
PC1	Fa0	192.168.1.20	255.255.255.0	192.168.1.1	Client LAN
PC2	Fa0	192.168.2.10	255.255.255.0	192.168.2.1	Direct to router

4. Step-by-Step Configuration

4.1 Router0 – Both Interfaces

```
Router>enable
Router#configure terminal
Router(config)#hostname Router0

! LAN interface — facing Switch0
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit

! PC2 interface — direct connection
Router0(config)#interface GigabitEthernet0/1
Router0(config-if)#ip address 192.168.2.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit

Router0(config)#exit
Router0#write memory
```

4.2 PC Static IP Configuration

PC0 – IP	192.168.1.10
PC0 – Mask	255.255.255.0
PC0 – Gateway	192.168.1.1
PC1 – IP	192.168.1.20
PC1 – Mask	255.255.255.0
PC1 – Gateway	192.168.1.1
PC2 – IP	192.168.2.10
PC2 – Mask	255.255.255.0
PC2 – Gateway	192.168.2.1

5. Verification

```
PC0>ping 192.168.1.20    ! PC0 to PC1 (same subnet via switch)
PC0>ping 192.168.2.10    ! PC0 to PC2 (cross-subnet via router)
PC2>ping 192.168.1.10    ! PC2 to PC0 (verify return path)

Router0#show ip interface brief  ! Gig0/0 and Gig0/1 must be up/up
Router0#show ip route            ! C 192.168.1.0 and C 192.168.2.0
```

6. Key Takeaways

- Two router interfaces automatically create two separate subnets.
- PC0 and PC1 share 192.168.1.0/24 — the switch handles their communication at Layer 2.
- PC2 is on 192.168.2.0/24 — requires the router to forward packets.
- Connected routes appear in the routing table automatically when interfaces are up.
- The 2960 switch needs no IP address or routing configuration for basic Layer 2 forwarding.